

Game Theory-Assignment1

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Exercise 1

(1)

For the action profile (N, S) to be a Nash equilibrium, neither player should have an incentive to deviate unilaterally.

For player 1, given that player 2 chooses S, choosing N must yield at least as much profit as choosing S:

$$a_{21} \geq a_{22}.$$

For player 2, given that player 1 chooses N, choosing S must yield at least as much profit as choosing N:

$$b_{12} \geq b_{11}.$$

Thus, (N, S) is a Nash equilibrium if and only if

$$a_{21} \geq a_{22}, \quad b_{12} \geq b_{11}.$$

Similarly, for the action profile (S, N) to be a Nash equilibrium, player 1 must not benefit from deviating when player 2 chooses N, and player 2 must not benefit from deviating when player 1 chooses S. These conditions are

$$a_{12} \geq a_{11}, \quad b_{21} \geq b_{22}.$$

Taking into account the assumption $a_{12} > a_{11}$ and $b_{21} > b_{11}$, both (N, S) and (S, N) are Nash equilibria if and only if

$$a_{12} \geq a_{11}, \quad a_{21} \geq a_{22}, \quad b_{12} \geq b_{11}, \quad b_{21} \geq b_{22}.$$

(2)

For (N, N) to be a Nash equilibrium, the following conditions must hold:

$$a_{11} \geq a_{12}, \quad b_{11} \geq b_{21}.$$

However, these inequalities contradict the assumption $a_{12} > a_{11}$ and $b_{21} > b_{11}$. Therefore, (N, N) cannot be a Nash equilibrium.

For (S, S) to be a Nash equilibrium, neither player should gain by deviating:

$$a_{22} \geq a_{21}, \quad b_{22} \geq b_{12}.$$

Since (N, N) cannot be a Nash equilibrium under the given assumptions, (N, N) and (S, S) cannot both be Nash equilibria.

Exercise 2

(1)

Consider each pure strategy profile in turn.

If (NA, D) is played, the government obtains payoff t by not auditing and $t - c$ by auditing. Since $t > c$, auditing yields a higher payoff, so the government has an incentive to deviate.

If (A, ND) is played, the taxpayer obtains payoff $1 - t - f$ and would strictly prefer declaring income, yielding payoff $1 - t$.

If (A, D) is played, the government obtains payoff $t - c$ but would prefer not auditing, which yields payoff t .

If (NA, ND) is played, the government obtains payoff 0 but would prefer auditing, yielding payoff $t + f - c > 0$.

Hence, no pure strategy profile can be a Nash equilibrium.

(2)

Suppose the government audits with probability $p_a \in [0, 1]$.

If the taxpayer declares income, the payoff is

$$U(D) = 1 - t.$$

If the taxpayer does not declare income, the expected payoff is

$$U(ND) = (1 - p_a) \cdot 1 + p_a(1 - t - f) = 1 - p_a(t + f).$$

The taxpayer prefers to declare income if and only if

$$1 - t \geq 1 - p_a(t + f),$$

which is equivalent to

$$p_a \geq \frac{t}{t + f}.$$

(3)

Suppose the taxpayer declares income with probability $p_d \in [0, 1]$.

If the government audits, its expected payoff is

$$U(A) = p_d(t - c) + (1 - p_d)(t + f - c) = t + f - c - p_d f.$$

If the government does not audit, its expected payoff is

$$U(NA) = p_d t.$$

The government prefers auditing if and only if

$$t + f - c - p_d f \geq p_d t,$$

which simplifies to

$$p_d \leq 1 - \frac{c}{t + f}.$$

(4)

Let

$$p_d = 1 - \frac{c}{t+f}, \quad p_a = \frac{t}{t+f}.$$

Under these probabilities, the taxpayer is indifferent between declaring and not declaring income, since

$$U(D) = 1 - t = U(ND).$$

Similarly, the government is indifferent between auditing and not auditing, since

$$U(A) = U(NA).$$

Therefore, neither player has an incentive to deviate from these mixed strategies, and the pair (p_d, p_a) constitutes a mixed-strategy Nash equilibrium.

Homework #1

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Exercise	weight	grade (0..4)
1.1	17.5%	2
1.2	17.5%	4
2.1	15%	4
2.2	15%	4
2.3	15%	4
2.4	20%	4

Grades

4 – correct

3 – mostly correct

2 – half-and-half

1 – mostly

incorrect 0 – not

done